

# Lead Router

## Standard Operating Procedure

GTM Operations

Google Sheets + Apps Script + n8n

## System Overview

The Lead Router is an automated outreach lead qualification and distribution system built for the client's GTM operations. It filters leads from a master database, qualifies them against configurable rules, removes blocklisted contacts, and pushes them to outreach campaigns on Instantly (email) and HeyReach (LinkedIn).

The system runs entirely from a Google Sheet with an Apps Script backend and an n8n workflow for campaign delivery.

## Architecture

Google Sheet (Master DB + Control + Filtered + Blocklist + Sent Log) ↓ Apps Script (filter engine + blocklist check + webhook trigger) ↓ n8n Webhook (receives trigger) ↓ n8n calls Apps Script doGet → gets filtered leads as JSON ↓ n8n routes by platform → Instantly API / HeyReach API ↓ n8n updates Master DB status + appends to Sent Log

## Spreadsheet Tabs

| Tab       | Purpose                                                                                                                   |
|-----------|---------------------------------------------------------------------------------------------------------------------------|
| Master DB | All leads. Source of truth. Contains outreach_channel, industry, seniority, region, city, state, status, Blocked columns. |
| Control   | Filter rules. Each row = one campaign. Defines which leads go where.                                                      |
| Filtered  | Output of each run. Shows matched leads with "Add to Blocklist" checkboxes. Cleared at the start of every run.            |
| Blocklist | Emails, domains, and LinkedIn URLs to exclude. Column A = entry, Column B = source tag.                                   |
| Sent Log  | Audit trail. n8n appends a row for every lead successfully sent to a campaign.                                            |

## Tab Structures

## Master DB — Key Columns

|  | Description                                                                          |           |        |
|--|--------------------------------------------------------------------------------------|-----------|--------|
|  | "Instantly" or "Heyreach" — determines which platform this lead belongs to           |           |        |
|  | Email address (required for Instantly leads)                                         |           |        |
|  | LinkedIn profile URL (required for HeyReach leads, also used for blacklist matching) |           |        |
|  | Full name of the lead                                                                |           |        |
|  | First and last name                                                                  |           |        |
|  | Company name                                                                         |           |        |
|  | One of 21 industry categories                                                        |           |        |
|  | Top Level / Medium Level / Other / Unknown                                           |           |        |
|  | Delhi / ROI / Out of India                                                           |           |        |
|  | City name                                                                            |           |        |
|  | State name                                                                           |           |        |
|  | Company domain                                                                       |           |        |
|  | Set by n8n after sending (e.g., "Added to Instantly \                                | camp_id \ | date") |
|  | TRUE/FALSE checkbox. TRUE = excluded from all filtering.                             |           |        |

## Control Tab — Columns

| Column            | Required? | Description                                                                          |
|-------------------|-----------|--------------------------------------------------------------------------------------|
| Campaign Name     | ■ Yes     | Name of this rule/campaign. Shows in sidebar and gets stamped on each lead in JSON.  |
| Outreach channel  | ■ Yes     | "Instantly" or "Heyreach". Determines platform AND which campaign ID column to read. |
| Industry category | Optional  | Multi-select chip dropdown. Leave blank = all industries.                            |
| Seniority level   | Optional  | Multi-select chip dropdown. Leave blank = all levels.                                |
| Region            | Optional  | Multi-select chip dropdown. Leave blank = all regions.                               |
| City              | Optional  | Multi-select chip dropdown. Leave blank = all cities.                                |
| State             | Optional  | Multi-select chip dropdown. Leave blank = all states.                                |

| Column                  | Required?             | Description                                          |
|-------------------------|-----------------------|------------------------------------------------------|
| Instantly Campaign Name | Reference             | Name of the Instantly campaign (for your reference). |
| Instantly Campaign ID   | Required if Instantly | The Instantly campaign UUID.                         |
| Heyreach Campaign Name  | Reference             | Name of the HeyReach campaign (for your reference).  |
| Heyreach Campaign ID    | Required if HeyReach  | The HeyReach campaign ID.                            |
| To Process              | ■ Yes                 | "Yes" or "No". Set to "No" to skip this rule.        |
| Notes                   | Optional              | Ignored by the script.                               |

#### Filter Logic:

- Within a column (e.g., Industry = "E-Commerce, Healthcare"): OR logic — lead matches if it's in ANY selected value.
- Across columns (e.g., Industry AND Region AND City): AND logic — lead must pass every filter that has values.
- Blank column = match all.

#### Filtered Tab

| Column              | Description                                            |
|---------------------|--------------------------------------------------------|
| Add to Blocklist    | Checkbox (column A). Tick to block, untick to unblock. |
| (remaining columns) | Mirror of Master DB headers.                           |

Blocked leads appear with pre-ticked checkboxes. Non-blocked leads have empty checkboxes. Only non-blocked leads are sent to campaigns.

#### Blocklist Tab

| Column    | Description                                                                   |
|-----------|-------------------------------------------------------------------------------|
| blocklist | The entry — email, domain, or LinkedIn URL. Script auto-detects type.         |
| source    | Blank = manually added. "marked_from_filtered" = added via the checkbox flow. |

#### Auto-detection rules:

- Contains "linkedin.com" → treated as LinkedIn URL
- Contains "@" → treated as email (only that email, not the whole domain)
- Everything else → treated as domain (blocks all leads from that domain)

## How to Run the System

### Menu Options

Open the Google Sheet → menu bar → **Lead Router**:

| Menu Item                | What it does                                                                                  |
|--------------------------|-----------------------------------------------------------------------------------------------|
| ■ Run Automation         | Filters leads, writes to Filtered tab, triggers n8n webhook to send to campaigns.             |
| ■ Dry run (preview only) | Filters leads, writes to Filtered tab for review. No status changes, no webhook.              |
| Move marked to Blocklist | Processes checkbox changes in Filtered tab. Also syncs manual blocklist entries to Master DB. |

## Running a Live Automation

1. Set up your rules in the Control tab (Campaign Name, Outreach channel, filters, campaign IDs, To Process = Yes).
2. Click **Lead Router** → **■ Run Automation**.
3. The sidebar opens showing:
  - Total available leads
  - New leads count
  - Instantly / HeyReach split
  - Per-rule breakdown with blocked count
  - Webhook status (success/failed/skipped)
4. n8n picks up the webhook, calls the Apps Script to get the leads, and sends them to the respective platforms.
5. n8n updates the status column in Master DB and appends to the Sent Log.

## Running a Dry Run

1. Click **Lead Router** → **■ Dry run (preview only)**.
2. Same filtering happens, leads are written to the Filtered tab for review.
3. No status changes in Master DB, no webhook triggered.
4. Review the Filtered tab — blocked leads appear with pre-ticked checkboxes.
5. If satisfied, run the live automation.

## Blocking Leads

### From the Filtered tab (after a dry run or live run):

1. Tick the "Add to Blocklist" checkbox for any lead you want to block.
2. Click **Lead Router** → **Move marked to Blocklist**.
3. The lead's defaultProfileUrl is added to the Blocklist tab.
4. The lead's Blocked column in Master DB is set to TRUE.
5. The lead won't appear in future runs.

### Unblocking a lead:

1. In the Filtered tab, untick the checkbox of a pre-ticked (blocked) lead.
2. Click **Lead Router** → **Move marked to Blocklist**.
3. The entry is removed from the Blocklist tab.
4. The Blocked column in Master DB is set to FALSE.
5. The lead becomes eligible again.

### Manually blocking (without Filtered tab):

1. Go to the Blocklist tab.
2. Add the email, domain, or LinkedIn URL in Column A. Leave Column B blank.
3. Click **Lead Router** → **Move marked to Blocklist** (this also syncs manual entries to Master DB).
4. Alternatively, directly tick the Blocked checkbox in Master DB — this also excludes the lead.

### Blocking an entire domain:

1. Go to the Blocklist tab.
  2. Type the domain (e.g., "competitor.com") in Column A.
  3. Run "Move marked to Blocklist" to sync.
  4. All leads with that domain (from domain\_name column or email) will be excluded.
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## How Filtering Works — Step by Step

When you click "Run Automation" or "Dry run", the script executes in this order:

1. **Clear Filtered tab** — wipes all previous results so you only see the current run.
  2. **Read Control tab** — finds all rules where To Process = "Yes". Reads the outreach channel to determine platform and campaign ID.
  3. **Read Master DB** — loads all leads, skips:
    - Phantom rows (no email AND no name)
    - Leads where Blocked = TRUE
  4. **Read Blocklist** — loads all entries from both columns. Auto-detects type (email/domain/LinkedIn URL).
  5. **For each rule:**
    - a. **Apply filters** — matches leads by outreach\_channel, industry, seniority, region, city, state. (AND across columns, OR within each column. Blank = match all.)
    - b. **Status check (Instantly only)** — for Instantly rules, removes leads where the status column has any value other than blank or "Filtered". HeyReach rules skip this check entirely (HeyReach handles dedup at the campaign level).
    - c. **Blocklist check** — separates leads into clean (not blocklisted) and blocked (in blocklist).
    - d. **Write to Filtered tab** — clean leads get unchecked checkbox, blocked leads get pre-ticked checkbox.
    - e. **Build JSON** — only clean leads go into the JSON output for n8n.
  6. **Mark status (live run only)** — currently disabled. n8n handles all status writes.
  7. **Trigger webhook (live run only)** — sends a POST to the n8n webhook URL.
  8. **Show sidebar** — displays results.
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## n8n Workflow

**Workflow name:** Lead Router - Adding leads to Instantly or Heyreach

**Webhook URL:** <https://n8n.aifyy.com/webhook/add-filtered-leads-to-campaigns>

### Flow

1. **Webhook** — receives the trigger from Apps Script.
2. **Fetch Filtered Leads** — HTTP GET to the Apps Script web app URL. Returns JSON with all filtered leads.

3. **Has New Leads?** — checks if totalLeads > 0.
4. **Split Leads** — splits the flat leads array into individual items.
5. **Route by Platform** — checks campaign\_platform on each lead. "Instantly" → Instantly branch, "HeyReach" → HeyReach branch.
6. **Send to Instantly** — HTTP POST to Instantly API with the lead data.
7. **Send to HeyReach** — HTTP POST to HeyReach API with the lead data.
8. **Update Master DB** — writes status like "Added to Instantly | camp\_id | date" via Google Sheets node.
9. **Append to Sent Log** — adds a row to the Sent Log tab.

## Important Notes

- The n8n workflow must be **Active** (green toggle in top-right) for the webhook to work in production.
- The webhook node must be set to **Production** mode, not Test mode.
- After updating the Apps Script, you must **publish a new web app version** (Deploy → Manage deployments → Edit → New version → Deploy) for doGet to use the updated code. The menu buttons always use the latest saved code, but the web app URL is pinned to a specific deployment version.

## JSON Output Structure

The Apps Script returns this JSON when n8n calls doGet:

```
{
  "success": true,
  "totalLeads": 20,
  "totalRules": 2,
  "totalMasterRows": 1726,
  "ruleSummary": [
    {
      "name": "Healthcare - Delhi - Email",
      "platform": "Instantly",
      "newLeads": 15,
      "blocked": 2
    },
    {
      "name": "Consumer Electronics - Delhi",
      "platform": "HeyReach",
      "newLeads": 5,
      "blocked": 1
    }
  ],
  "leads": [
    {
      "email": "person@company.com",
      "linkedin_url": "https://linkedin.com/in/person",
      "full_name": "Person Name",
      "first_name": "Person",
      "last_name": "Name",
      "company": "Company",
      "title": "Marketing Manager",
      "industry": "Healthcare",
      "domain": "company.com",
      "city": "Delhi",
      "state": "Delhi",
      "country": "India",
      "region": "Delhi",
      "seniority_level": "Top Level",
      "outreach_channel": "Instantly",
      "campaign_id": "e9516a18-9fe4-43ca-93b9-a04381084ffc",
      "campaign_name": "demo_campaign",
      "campaign_platform": "Instantly",
      "rule_name": "Healthcare - Delhi - Email"
    }
  ]
}
```

## Troubleshooting

### "No active filter rules found"

- Check that the Control tab has at least one row with To Process = "Yes".
- Verify the outreach channel is exactly "Instantly" or "Heyreach" (case-insensitive but must be one of these two).
- Make sure the Campaign Name is filled.
- Make sure the relevant Campaign ID column is filled.

### 0 leads matched but there should be matches

- Run a dry run first and check the Filtered tab.
- Verify the filter values in the Control tab exactly match the values in Master DB (e.g., "E-Commerce" vs "e-commerce" — the script handles case, but check for typos or extra spaces).
- Check if the leads are marked as Blocked = TRUE in Master DB.
- For Instantly rules, check if the leads already have a status value in the status column.

### n8n webhook not triggering

- Verify the workflow is Active in n8n (green toggle, top-right).
- Verify the Webhook node is set to Production mode, not Test.
- Check the webhook URL matches between the script and n8n.
- Check if your n8n hosting has bandwidth limits.

### doGet returns 0 leads but menu shows leads

- You need to publish a new web app version after updating the script. Deploy → Manage deployments → Edit → New version → Deploy.

### "Cannot freeze all visible rows" error

- This happens when the Filtered tab has only 1 row (just the header). The script handles this, but if you see it, republish the web app.

### Blocked leads still appearing

- Run "Move marked to Blocklist" to sync the blocklist to Master DB.
- Check that the Blocked column exists in Master DB with the exact header "Blocked".
- Verify the Blocked column has a TRUE/FALSE checkbox or the text "TRUE".

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## Setup Checklist (First Time)

1. Open the Google Sheet.
2. Verify the Master DB tab has these columns: outreach\_channel, email, defaultProfileUrl, fullName, industry\_category, seniority\_level, region, city, state, status, Blocked.
3. Create the Control tab with the required columns (or verify it exists).
4. Create the Blocklist tab by running Lead Router → Move marked to Blocklist (it auto-creates if missing).
5. Open the Apps Script editor, paste the script, save.
6. Deploy as web app (Deploy → New deployment → Web app → Execute as: Me → Who has access: Anyone → Deploy).
7. Copy the web app URL and paste it into the n8n "Fetch Filtered Leads" HTTP node.
8. Activate the n8n workflow.
9. Close and reopen the Google Sheet to see the Lead Router menu.
10. Run a dry run to verify everything works.

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## Key Design Decisions

- **No status marking from script.** The script never writes "Filtered" to the status column. n8n handles all status writes after successfully sending leads.
- **HeyReach skips status check.** HeyReach has built-in campaign-level dedup, so the script doesn't check the status column for HeyReach rules. Instantly rules still respect the status column.
- **Blocklist is two-way.** Tick a checkbox in the Filtered tab to block. Untick a pre-ticked checkbox to unblock. Both directions work in one button click.
- **Blocked column in Master DB is the fast gate.** Leads with Blocked = TRUE are excluded before any filtering. The Blocklist tab is the source of truth; the Blocked column mirrors it.
- **Both dry run and live run write to Filtered tab.** This lets you review exactly which leads will go out before committing.

- **One row in Control tab = one campaign.** To send the same leads to multiple campaigns, add multiple rows with the same filters but different campaign IDs.